

1 Agroforestry SmartAgroforst: Poplar biomass modelling

2 Thomas Kirschstein^{1,1,1,*}

3 Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

4 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
5 cascading

6 1. Introduction

7 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
8 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
9 biodiversity conservation, and rural income generation on the same hectare of land (Toens-
10 meier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such

*Corresponding author

Email address: thomas.kirschstein@ikts.fraunhofer.de (Thomas Kirschstein)

Preprint submitted to Journal Name

May 27, 2026

11 as poplar, willow, alder, and black locust can deliver substantial biomass yields while im-
12 proving soil quality and providing multiple ecosystem services compared to conventional
13 monocultures (Huber et al., 2018). Biomasses produced in AFS is considered as a sus-
14 tainable feedstock and can be used either energetically, chemically or materially (e.g. in
15 the building sector). The latter two options are generally preferred over the energetic
16 usage from a cleaner production perspective as biogenic carbon is stored in products
17 during their life span which contributes to climate change mitigation.

18 Despite substantial research on ecological advantages and pioneering initiatives to
19 establish AFS in practice, the share of AFS in agricultural production systems is still
20 small. E.g. in Germany about 1200 hectares of silvoarable AFS are established in 2025,
21 which accounts for about 0.02% of total agricultural land use [DEFAP]. One of the core
22 challenges in establishing AFS systems at large scale in practice is a lack of opportunities
23 to generate substantial revenues from woody biomasses of AFS. On the other side, the
24 rise of bioeconomy creates substantial demands for woody biomass and requires a reliable
25 and cost-efficient feedstock supply. As woody biomasses are bulky feedstocks with low
26 energy density, the design of cost-efficient supply chains is crucial to ensure the economic
27 viability of AFS-based value chains. This requires coordinated decisions on stand estab-
28 lishment, multi-cycle harvesting, and product cascading in spatially distributed supply
29 chains, which can be supported by integrated optimisation models that link stand-level
30 growth dynamics to regional supply chain design.

31 In the last two decades, a large body of research has analysed biomass supply chains
32 for bioenergy and biorefineries using mathematical programming models, often in the
33 form of mixed-integer linear programming (MILP) (Zahraee et al., 2020; Sharma et al.,
34 2013). These models typically optimise network configuration, feedstock sourcing, and
35 logistics under cost or profit objectives. They often treat biomass availability as an exoge-
36 nous input or rely on simplified agricultural production decisions. Moreover, downstream
37 processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants
38 consuming all produced biomasses. Recent work has started to integrate operational de-
39 tails such as various biomass types, biomass growth models and regeneration processes
40 in multi-period supply chain models.

41 The present paper contributes to this literature in four ways. First, we propose

42 a novel MILP formulation for AFS-based biomass supply chain design that explicitly
43 links establishment decisions and consecutive harvest decisions, ensuring that harvesting
44 decisions at each site are consistent with minimum and maximum rotation ages. Second,
45 we embed allometric yield functions for different biomass types generating age-specific
46 yields for different woody biomass types (like stems, branches, or leaves and twigs). Third,
47 these different woody biomass types can be processed in specific downstream industries,
48 like chemical or paper industry as well as energy sector. The model aims to provide
49 strategic decision support for establishing regional AFS-based bioeconomy ecosystems
50 connecting multiple industries and product markets.

51 The remainder of the paper is structured as follows. Section 2 reviews the relevant lit-
52 erature on agroforestry biomass systems, allometric biomass modelling, and biomass sup-
53 ply chain optimisation. Section 3 summarises the MILP formulation for the agroforestry
54 supply chain with age-lag constraints. Section ?? outlines the planned computational
55 experiments for the case study and discusses expected insights for cleaner production
56 strategies.

57 2. Literature review

58 Agroforestry biomass systems, stand-level growth models, and biomass supply chain
59 optimisation have each been studied extensively, but they are rarely combined in an inte-
60 grated framework that links age-dependent stand dynamics to regional product-cascading
61 value chains (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). This sec-
62 tion reviews (i) agroforestry systems with emphasis on system types, economics, and the
63 cascading use of biomass as the primary revenue driver; (ii) biomass growth models at
64 tree and stand level with explicit attention to the allometric–diameter–stand-biomass
65 pathway; and (iii) biomass supply chain design using mathematical programming. The
66 section concludes by identifying the modelling gap addressed in this paper.

67 2.1. Agroforestry systems

68 Agroforestry deliberately integrates woody perennials with crops and/or livestock on
69 the same management unit, with the aim of enhancing productivity, profitability, and
70 environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three

71 distinct system types can be distinguished based on the spatial arrangement of trees and
72 the rotation length at which woody biomass is harvested.

73 **Short-rotation coppice (SRC)** is defined as a monoculture plantation of fast-
74 growing woody species covering the entire management unit, with planting densities
75 of 8,000–20,000 cuttings $\cdot\text{ha}^{-1}$. After each harvest the root system regenerates without
76 replanting. Rotation cycles are typically 3–7~years and total stand lifetime is 20–25~years
77 (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense,
78 but it is helpful to benchmark biomass production.

79 **Short-rotation agroforestry systems (SRAFS)** integrate SRC-type tree strips
80 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an otherwise
81 arable field in an alley-cropping arrangement. Crop production continues in the inter-
82 row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover
83 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are
84 comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light
85 and water availability than interior SRC trees. This boundary effect raises individual-tree
86 biomass accumulation but also intensifies competition with adjacent crop rows (Huber
87 et al., 2018).

88 **Long-rotation agroforestry systems (LRAFS)** use the same alley-cropping spa-
89 tial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al.,
90 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem di-
91 ameters and the proportion of high-value timber increases markedly, enabling premium
92 material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-
93 rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry,
94 oak, or high-grade poplar clones) are planted.

95 SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age
96 is hard to define and, mor importantly, the intended use of AFS biomass is not always
97 clear and can also change over time. Nonetheless, when establishing AFS, the structure
98 of AFS and its intended use after harvest drives planting decisions and, subsequently,
99 economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as
100 separate AFS systems, but the interplay of intended use, establishment, and harvesting
101 decisions is analysed.

2.1.1. *Economic rationale of landowners*

The decision of a landowner to establish an AFS involves weighing substantial up-front costs and long-term opportunity costs against a diversified revenue stream. On the **cost side**, establishment requires site preparation, planting material, and planting labour, typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude, 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping, transport to roadside) add approximately 200–400 €/ha per rotation depending on stand density and equipment availability (Testa et al., 2014; Spinelli et al., 2009). During the rotation, maintenance cost of AFS arise on a yearly basis. Next to expenses for establishment, maintenance and harvesting, opportunity cost arise as no revenues from crops are generated on tree strips. In regions with fertile soils and high cereal prices, yearly crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves et al., 2007). Thus, opportunity cost are often the dominant cost driver in SRAFS profitability analyses.

On the **benefit side**, the primary revenue of AFS is the sale of harvested woody biomass, whose magnitude depends critically on rotation length, species, and site productivity. For poplar-based SRAFS in Central Europe, typical yields range from 10 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018; Niemczyk and Thomas, 2021). Additionally, positive crop-yield effects in the inter-row strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil erosion and evapotranspiration; they provide partial shading that moderates heat stress during drought years; and their root systems may improve water retention in the topsoil (Graves et al., 2007; Grünewald et al., 2007). Empirical measurements in Central European alley-cropping trials indicate that these microclimate effects can increase cereal yields in adjacent strips by 5–15% relative to open-field controls, partially or fully compensating the loss of arable area occupied by the tree strips (Graves et al., 2007). Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to agri-environment subsidies and carbon credits (Toensmeier, 2017; Faasch and Patenaude, 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS

133 biomass. Thereby, revenues are different for the types of biomass and in which value
 134 chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power
 135 generation, or anaerobic digestion, which typically creates low prices per tonne of biomass.
 136 Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also
 137 be used as pulp and paper grade material as well as feedstock for chemical industry
 138 or as timber achieving higher prices (Lamers et al., 2011; De Meyer et al., 2014). Thus,
 139 fostering AFS establishment by maximizing economic returns from AFS biomass requires
 140 a strategic approach to product cascading that allocates different biomass fractions to
 141 the highest-value applications, while ensuring that the rotation length is aligned with
 142 the growth dynamics of the stand and the market demand for different biomass types
 143 (Lamers et al., 2011).

144 Thus, for making educated decisions on AFS establishment and harvesting, the ex-
 145 pected biomass quantities at harvest for stem, branches, and residues need to be esti-
 146 mated. Thereby, the shares of the biomass types change with stand age. E.g.~the share
 147 of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018;
 148 Testa et al., 2014). Therefore, in Section 2.2 growth modelling approaches are reviewed
 149 that capture the age-dependent growth dynamics of different biomass types in AFS.

150 2.2. Biomass growth models

151 Total biomass on tree level is commonly modeled using allometric functions that
 152 relate tree characteristics like stem diameter or height to total biomass weight. There are
 153 numerous allometric models in the literature, often specific to species, site conditions, and
 154 stand age. For fast-growing species used in SRAFS (like poplar and willow), allometric
 155 functions of the power-law form are used [Reference]. I.e., dry biomass M^{sh} (kg) of a
 156 shoot is approximated by

$$M^{sh} = b^0 \cdot SBD^{b^1} \quad (1)$$

157 where SBD is the shoot's stem base diameter (in cm) measured 10 cm above ground.
 158 Parameters b^0 and b^1 are species-specific (Huber et al., 2016). In the following it is
 159 assumed that each tree is pruned to one dominant shoot. For SRAFS species in Southern
 160 Germany, Huber et al. (2018) reports that most trees produce not more than 2 shoots
 161 and Huber et al. (2016) report a common exponent $b_1 = 2.603$ for black alder, black

Table 1: Notation for the biomass growth models.

Symbol	Description
M^{sh}	Dry biomass (kg) per shoot
$M^{sh}(t)$	Mean shoot biomass (kg) at stand age t
M^{sl}	aboveground dry biomass per ha (t)
$M^{sl}(t)$	aboveground dry biomass per ha (t) at stand age t
SBD	Stem base diameter (cm) measured 10 cm above ground
b^0, b^1	Allometric coefficients for shoot biomass estimation
$SBD(t)$	Mean stem base diameter (cm) at stand age t
a^∞	Asymptotic stand-level biomass (t dry matter/ha)
b^∞	Asymptotic mean shoot biomass ($= d^\infty \cdot b^0$)
d^∞	Asymptotic mean stem base diameter (cm)
r	Yearly diameter growth rate
b'^1	Effective shoot-biomass growth rate ($= r \cdot b^1$)
τ	Age at inflection point of diameter growth curve (yr)
$N(t)$	Stand density (trees/ha) at age t
$\bar{n}_{sh}(t)$	Mean number of shoots per tree at age t
$h_p(t)$	Gompertz helper function for biomass component p
$f_p(t)$	Normalised share of component p in total AGB at age t
a^p	Asymptotic biomass of component p
b^p	Growth rate of biomass component p
τ_p	Inflection age of component p (yr)

locust, poplar (Max~3), and willow (Inger), while b^0 ranges from 0.025 to 0.041, reflecting differences in wood density and branching structure.

Allometric rules like (1) generate estimates of biomass quantities for a given shoot, i.e. they can be used to support operational harvest decisions. For planning on a strategic level, it is necessary to estimate biomass development over time, i.e. an SBD growth model. The mean stem base diameter $SBD(t)$ of coppice shoots increases rapidly in the first growing seasons as shoots exploit stored root carbohydrates, then decelerates as intra-stand competition for light and water intensifies. Such trajectories can be described by sigmoidal growth functions. Empirical evidence from poplar SRC and SRAFS trials indicates that the SBD growth curve exhibits maximal growth rate (i.e., inflection point) at a relatively early stand age (2–4 years) (Barrio-Anta et al., 2008; Niemczyk and

173 Thomas, 2021). Therefore, a Gompertz model for $SBD(t)$ can be formulated as

$$SBD(t) = d^\infty \cdot e^{-e^{-r \cdot (t-\tau)}}, \quad (2)$$

174 where d^∞ (cm) is the asymptotic mean diameter, r controls the rate of yearly diameter
175 increment, and τ (in years) is the age at the inflection point of the diameter curve. Given
176 $SBD(t)$ from (2), the mean shoot biomass growth follows directly from the allometric
177 equation (1):

$$M^{sh}(t) = b^0 \cdot SBD(t)^{b^1} = d^\infty \cdot b^0 \cdot e^{-e^{-r \cdot b^1 \cdot (t-\tau)}}. \quad (3)$$

178 Thereby, $b^\infty = d^\infty \cdot b^0$ denotes the asymptotic mean shoot biomass, and the effective
179 growth rate of shoot biomass is $b'^1 = r \cdot b^1$, which is amplified relative to the biomass
180 growth rate r by the allometric exponent b^1 .

181 Total aboveground biomass quantities on stand level $M^{sl}(t)$ (in t per ha) is then
182 obtained by aggregating individual shoot biomasses over all trees and multiplying by
183 stand density $N(t)$ (in trees per ha):

$$M^{sl}(t) = N(t) \cdot \bar{n}^{sh}(t) \cdot M^{sh}(t), \quad (4)$$

184 where $\bar{n}^{shoot}$ is the mean number of shoots per tree (Huber et al., 2018). Both $\bar{n}^{sh}(t)$
185 and $N(t)$ may change over time. The number of shoots is often controlled in tree care
186 processes and reduced to 1 or 2 shoots per tree (Huber et al., 2018). Stand level density
187 $N(t)$ is initially defined when the AFS is established and often declines somewhat over
188 time, but usually saturates quickly [REFERENCE]. For the product $M^{sl}(t)$ in (4) a
189 right-skewed sigmoidal shape with an inflection occurring at a relatively low fraction
190 of the asymptote can be assumed which can be captured in a Gompertz model as well
191 (Niemczyk and Thomas, 2021; Testa et al., 2014). Assuming constant tree density and
192 an average shoot number of 1, $M^{sl}(t)$ can be simplified as

$$M^{sl}(t) = a^\infty \cdot e^{-e^{-\bar{b}^1 \cdot (t-\tau)}}. \quad (5)$$

193 where $a^\infty = N \cdot \bar{n}^{sh} \cdot d^\infty \cdot b^0$ is the asymptotic stand-level biomass. For poplar clone
194 Max~3 under SRAFS conditions in Central Europe, representative parameter values are
195 $d^\infty = 9.5$ cm, $r = 0.55$, and $\tau = 2.5$ years, yielding a mean SBD of approximately 4–5
196 cm at age 3 and 7–9 cm at age 7 which is consistent with the diameter ranges reported

by Huber et al. (2018) and Barrio-Anta et al. (2008). For total aboveground biomass of poplar clone Max 3 under SRAFS management, the Gompertz model (5) is parameterised as $a^\infty = 100$ t of dry biomass per ha, $\tilde{b}^1 = 0.32$ per year, and $\tau = 4.0$ years, yielding a growth peak at approximately age 4 and a total aboveground biomass of roughly 65 t of dry biomass per ha at age 10 which is consistent with the empirical reference range of 50–135 t per ha reported for comparable poplar systems in the literature (Huber et al., 2018; Niemczyk and Thomas, 2021).

Yet, for the purpose of modelling AFS supply chains, the three biomass components merchantable stem, structural branches, and fine residues need to be distinguished. Thus, the total biomass growth model (5) needs to be adjusted those types. Thereby, growth characteristics are different each type and, thus, require parameterization. Thereby, the differentiation between the merchantable stem and branch depends on the requirements defined by downstream processes. E.g. for chemical production, the merchantable stem is defined as the portion of the tree with a certain minimum diameter (e.g. 10 cm), while branches and residues are defined as the portion of the tree that cannot be chipped into such large chips and thus require different processing or are used for lower-value applications. The share of these components in total aboveground biomass changes with stand age, which has important implications for product cascading decisions in AFS-based supply chains.

Empirical studies show that the stem fraction rises from roughly 55% of AGB at age~4 to over 75% at age~10, while branch and residue shares decline as stands mature (Huber et al., 2018; Testa et al., 2014, Jha (2018)). This age-dependence is biologically driven by the increasing dominance of secondary stem growth (radial increment) relative to primary shoot elongation and branching in later stand development stages. To capture these dynamics analytically, each component share $f_p(t)$ is represented as a normalised Gompertz function:

$$h_p(t) = a_p \cdot e^{-e^{-b_p \cdot (t - \tau_p)}} \quad (6)$$

$$f_p(t) = \frac{h_p(t)}{\sum_{p'} h_{p'}(t)}. \quad (7)$$

For Max~3 the stem helper is parameterised with a high asymptote ($a_{\text{stem}} = 1.0$) and a relatively late inflection ($\tau_{\text{stem}} = 4.0$ years), whereas the branch uses a lower asymptote

($a_{\text{branch}} = 0.7$) and an earlier inflection ($\tau_{\text{branch}} = 3.0$ years), and the residue helper saturates earliest ($a_{\text{rest}} = 0.1$, $\tau_{\text{rest}} = 2.0$ years). This parameterisation reproduces the observed shift towards stem dominance with age and is consistent with the empirical component fractions compiled across poplar agroforestry and SRC systems spanning ages 6–13 years, as illustrated in Figure ??.

2.3. Biomass supply chain design

Biomass supply chain design problems have been extensively studied in literature (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). Most models jointly optimise facility locations and capacities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost or profit objectives, sometimes augmented by greenhouse gas emission indicators (Ekşioğlu et al., 2009; Arabi et al., 2019). A key methodological feature of the more advanced formulations is the integration of long-term network design with multi-period logistics, enabling simultaneous optimisation of infrastructure and operational decisions across seasonal or annual time steps (Ekşioğlu et al., 2009; Zahraee et al., 2020).

Among generic multi-product frameworks, OPTIMASS explicitly models product quality cascades by defining a hierarchy of biomass classes and allowing higher-quality products to satisfy lower-quality demands, while the t-OPTIMASS extension incorporates biomass growth and regeneration over multiple periods (De Meyer et al., 2014, 2016). However, in virtually all existing formulations biomass availability is treated as an exogenous time-series input: stand age structure, rotation decisions, and coppice cycles are not represented endogenously, so that trade-offs between rotation length, stand management, and supply chain configuration cannot be explored in a unified model (De Meyer et al., 2016; Lamers et al., 2011). This gap is particularly pronounced for agroforestry systems with repeated coppice cycles, where establishment timing, consecutive harvest decisions, and age-dependent biomass quality jointly determine both the feasible supply profile and the achievable product cascade.

The MILP formulation developed in this paper addresses this gap by endogenising stand age dynamics through binary age-lag variables, coupling the component-specific yield coefficients η_{pa} derived from the Gompertz growth models above with supply chain flow variables, and embedding a product quality cascade that allocates stem, branch,

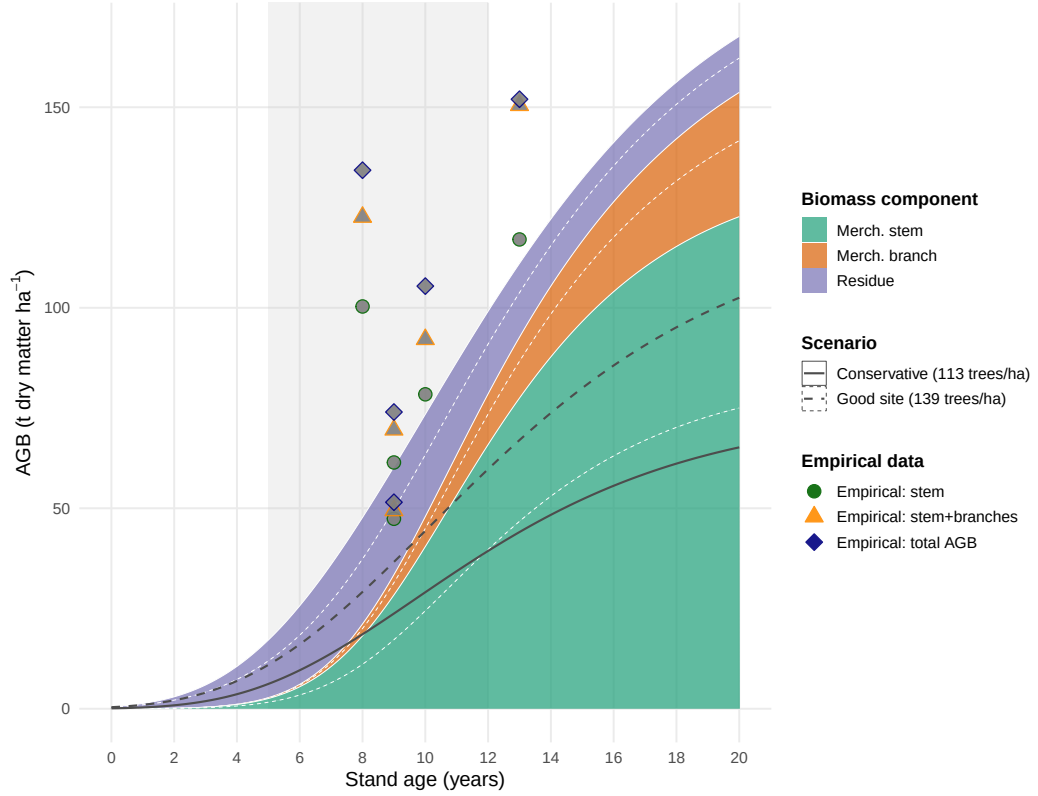


Figure 1: Stand-level aboveground biomass components for poplar under AFS management, modelled via a density-dependent Gompertz stand model with linear compartment fractions and logistic diameter-class merchantability. Shaded stacked areas: merchantable stem ($d > 15$ cm, green), merchantable branches ($d > 7$ cm, orange), and residues / fine material (purple). Both the conservative (solid, 113 trees/ha) and good-site (dashed, 139 trees/ha) scenarios are shown. Empirical aboveground biomass observations compiled from Jha (2018), Civitarese (2019), and Niemczyk (2021) are overlaid as symbols (filled circles: stem cumulative; triangles: stem + branches; diamonds: total AGB). Grey band: typical medium-rotation harvest window (5–12 years).

and residue fractions to chemical, pulp, and energy markets within a single optimisation model.

2.4. Biomass supply chain design

The design and planning of biomass supply chains has attracted substantial research attention over the past two decades, driven by the need to ensure cost-efficient and sustainable feedstock provision for bioenergy and biorefinery applications. Several systematic reviews provide comprehensive overviews of this literature. Cambero and Sowlati (2014) survey economic, social, and environmental assessment and optimisation studies for forest biomass supply chains, identifying MILP as the dominant modelling approach and noting a growing trend towards multi-objective formulations integrating life cycle assessment.

Yue et al. (2014) review biomass-to-bioenergy and biofuel supply chain models with a focus on multi-scale modelling, uncertainty treatment, and sustainability metrics, and conclude that the coupling of upstream feedstock dynamics with downstream network design remains an underexplored area. Malladi and Sowlati (2018) survey logistics-specific features of biomass supply chain models—seasonal availability, moisture content degradation, preprocessing steps—and highlight that most models treat biomass as a homogeneous commodity, neglecting quality differentiation. Zahraee et al. (2020) provide a recent meta-review of 140+ studies, confirming that cost minimisation is still the dominant objective, and that multi-product cascading formulations are rare.

The structure of biomass supply chain models typically encompasses three decision layers: (i) strategic network design (facility location, capacity sizing, technology selection), (ii) tactical planning (harvest scheduling, seasonal flows, inventory), and (iii) operational routing. The following paragraphs discuss key individual contributions and their specific methodological advances.

Ekşioğlu et al. (2009) develop one of the earliest large-scale MILP models for biomass-to-biorefinery supply chain design, integrating facility location, capacity sizing, and multimodal transport for lignocellulosic feedstocks. Their model minimises total system cost and introduces binary facility-opening variables that have since become standard in the field. A key limitation, however, is that biomass supply is treated as an exogenous input, so feedstock availability is not endogenised as a function of land-use or growth decisions.

287 Mansoornejad et al. (2013) extend the strategic design perspective to forest biorefiner-
 288 ies by integrating product portfolio design with supply chain configuration in a single
 289 MILP. Their model explicitly accounts for product conversion efficiencies along differ-
 290 ent processing pathways, enabling joint optimisation of process technology selection and
 291 logistics.

292 De Meyer et al. (2014) introduce the OPTIMASS framework, a generic MILP for
 293 biomass-based supply chain optimisation that explicitly models product quality cas-
 294 cades: higher-quality biomass classes can substitute for lower-quality demands down-
 295 stream, whereas the reverse is infeasible. This cascade hierarchy, formalised through
 296 ordered product sets, is particularly relevant for woody biomass where stem, branch,
 297 and residue fractions command different market prices. The companion t-OPTIMASS
 298 extension (De Meyer et al., 2016) incorporates temporal dynamics by explicitly repre-
 299 senting biomass growth and regeneration over multiple periods, enabling simultaneous
 300 optimisation of harvest timing and network flows. However, in both OPTIMASS for-
 301 mulations the growth dynamics are represented as exogenous time-series rather than as
 302 age-structured stand models, so rotation-length decisions and age-dependent biomass
 303 quality are not endogenised. Arabi et al. (2019) demonstrate multi-period MILP exten-
 304 sions for bio-based supply chains with production dynamics, but focus on microalgae
 305 rather than woody biomass and do not address product quality cascades.

306 Sharma et al. (2013) identify supply-side uncertainty (yield variability, seasonal avail-
 307 ability, moisture content) as the most critical source of risk in biomass supply chains.
 308 In response, a strand of literature has developed stochastic and robust optimisation ex-
 309 tensions. Zahraee et al. (2020) document that two-stage stochastic MILP and robust
 310 optimisation approaches now account for roughly 30% of the modelling literature, with
 311 biomass yield uncertainty and price volatility as the dominant stochastic parameters.

312 Biomass supply chain research focusing specifically on agroforestry or SRC contexts
 313 is comparatively sparse. Espinoza et al. (2017) note that most supply chain models as-
 314 sume continuous feedstock availability and do not represent the periodic harvest cycles
 315 characteristic of coppice systems. Lamers et al. (2011) analyse feedstock design principles
 316 for lignocellulosic supply chains and argue that the allocation of different biomass frac-
 317 tions to valorisation pathways—product cascading—is a key lever for economic viability,

yet is rarely optimised jointly with network design decisions. The model developed in Section~3 of this paper directly addresses this gap by endogenising stand age dynamics through binary age-lag variables and coupling component-specific yield coefficients with a product quality cascade.

Table~2 summarises the key modelling features of the reviewed contributions.

Table 2: Key modelling features of selected biomass supply chain design studies.

Study	Objective	Multi-product	Quality cascade	Age dynamics	Uncertainty	Level
Eksioglu et al. (2009)	Cost min.	—	—	—	—	Strategic
Sharma & Taaffe (2013)	Cost min.	Partial	—	—	—	Strategic/Tactical
Mansoornejad et al. (2013)	Profit max.	✓	—	—	—	Strategic
Cambero & Sowlati (2014)	Cost/GHG/jobs	—	—	—	Stochastic	Strategic
Yue et al. (2014)	Cost/GHG	✓	—	—	—	Multi-scale
De Meyer et al. (2015)	Cost min.	✓	✓	Exogenous	—	Strategic/Tactical
De Meyer et al. (2016)	Cost min.	✓	✓	Exogenous TS	—	Multi-period
Lamers et al. (2015)	Cost/quality	✓	Partial	—	Scenario	Strategic
Espinoza et al. (2017)	Sustainability	✓	—	—	Scenario	Strategic
Malladi & Sowlati (2018)	Cost min.	Partial	—	—	—	Tactical/Oper.
Arabi et al. (2019)	Profit max.	✓	—	Exogenous	—	Multi-period
Zahraee et al. (2020)	Multi-obj.	✓	—	—	Stochastic	Strategic/Tactical
This paper	NPV max.	✓	✓	Endogenous	Scenario	Strategic/Multi-period

3. MILP Model for Agroforestry Supply Chain Design

Building on the biomass growth models introduced in Section 2.2 and the supply chain design literature reviewed in Section 2.3, this section describes the AFS supply chain design problem (AFS-SCD), a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain. The model endogenises stand age dynamics through binary age-lag variables, couples the component-specific yield coefficients η_{pa} derived from the Gompertz growth models (5)–

(7) with supply chain flow variables, and assigns AFS biomass types (i.e., stem, branch, and residue fractions) to potential consumers in chemical, pulp and paper as well as energy industry.

3.1. Sets, Indices, and Time Structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{\text{stem}, \text{branches}, \text{residues}\}$, whose growth models are outlined above in (6)–(7). Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To model flexible harvest timing according to typical rotation structures of SRAFS and LRAFS, $A^{\min}$ and $A^{\max}$ define minimum and maximum stand ages. Thus, harvest are possible in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$. To consistently model establishment and consecutive harvest decisions, the following arc sets are defined: $\mathcal{S}^{\text{est}}$ defines potential establishments of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set $\mathcal{S}^{\text{harv}}$ summarizes all potential consecutive harvest periods within the age-lag bounds $A^{\min}$ and $A^{\max}$. I.e., if an AFS is established or harvested in period s and next harvest is in period t , arc (s, t) is selected. Finally, there is no subsequent harvest after period t , the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs are summarized in set $\mathcal{S}^{\text{term}}$. All arc sets are formally defined as follows:

$$\begin{aligned}\mathcal{S}^{\text{est}} &= \{(0, t) \mid t \in \{1, \dots, T^{\max} - A^{\min}\}\}, \\ \mathcal{S}^{\text{harv}} &= \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \dots, A^{\max}\}\}, \\ \mathcal{S}^{\text{term}} &= \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}, \\ \mathcal{S} &= \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}.\end{aligned}$$

Figure 2 illustrates this network structure of all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.

3.2. Decision Variables

Binary variables z_{ist} indicate whether an arc from set $(s, t) \in \mathcal{S}$ is selected for site i , indicating establishment, (consecutive) harvest, or termination. Taken together across all arcs, these variables trace a path of consecutive harvest cycles at each site over the planning horizon. Figure 3 illustrates a feasible path corresponding to the establishment

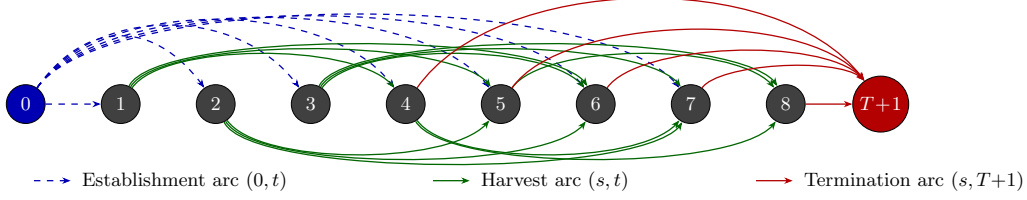


Figure 2: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $(0, t)$, $t \in \{1, \dots, T-1\}$. Solid green arcs above: harvest arcs (s, t) with odd s ; below: harvest arcs with even s . Solid red arcs: termination arcs $(s, T+1)$, $s \in \{A_{\min} + 1, \dots, T\}$.

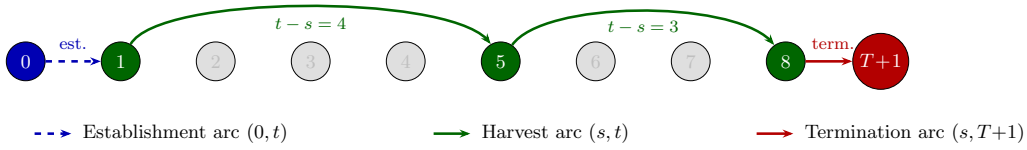


Figure 3: Example of a feasible path through the network for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$: establishment in period 1, harvests in periods 5 and 8, termination at $T + 1$.

354 of an AFS in period 1 and harvests in periods 5 and 8, whereby the harvest in $t = 8$ is
 355 final (and AFS is terminated, thus).

356 When a AFS is harvested in period t , continuous variables $Y_{ipt} \geq 0$ represent the
 357 harvest quantity (t) of product p at site i . Flow variables X_{ijpt} and $X_{jkpp't}$ represent
 358 transport of products from AFS sites to storage/pre-treatment facilities and from there
 359 to consumers, respectively, while S_{jpt} captures end-of-period inventory at facility j .

360 Table 3 summarises all sets, decision variables, and parameters used in the MILP
 361 formulation. Symbols shared with the growth model of Section 2.2 — in particular the
 362 yield coefficients η_{pa} , which are derived from the component-specific Gompertz functions
 363 (6)–(7) and the stand biomass model (5) — are listed in Table 1 in Section 2.2.

364 3.3. Objective Function

365 As discussed in Section 2.1, the economic attractiveness of AFS supply chains must be
 366 assessed from a global perspective taking into account all processes, activities, and partic-
 367 ipants involved in finally serving customer markets with bio-based feedstocks. Therefore,
 368 revenues for the different biomass fractions and must be weighed against costs occurring
 369 at all value-adding stages of the supply chains. To be precise, revenues created by selling

370 AFS-based feedstocks to different industrial consumers are counterweighted by establish-
 371 ment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms
 372 as well as biomass logistics companies. The model therefore maximises total contribution
 373 margin of all participants in the AFS supply chain over the planning horizon as follows:

$$\begin{aligned}
 \max Z = & \sum_{k \in \mathcal{K}, p \in \mathcal{P}} R_{kp} \cdot \sum_{j \in \mathcal{J}, (p', p) \in \mathcal{Q}, t \in \mathcal{T}} X_{jkp't} \\
 & - \sum_{i \in \mathcal{I}} c_i^{est} \cdot AREA_i \cdot \sum_{(0, t) \in \mathcal{S}^{est}} z_{i0t} \\
 & - \sum_{i \in \mathcal{I}} c_i^{opp} \cdot AREA_i \cdot \left(\sum_{(s, T+1) \in \mathcal{S}^{term}} (s+1) \cdot z_{is(T+1)} - \sum_{(0, t) \in \mathcal{S}^{est}} t \cdot z_{i0t} \right) \\
 & - \sum_{i \in \mathcal{I}} c_i^{main} \cdot (t-s) \cdot AREA_i \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
 & - \sum_{i \in \mathcal{I}} c_i^{harv} \cdot AREA_i \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
 & - \sum_{i \in \mathcal{I}, j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}} c_p^{tr-raw} \cdot d_{ij} \cdot X_{ijpt} \\
 & - \sum_{j \in \mathcal{J}, k \in \mathcal{K}, (p, p') \in \mathcal{Q}, t \in \mathcal{T}} c_{p'}^{tr-pre} \cdot d_{jk} \cdot X_{jkpp't} \\
 & - \sum_{j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}} c_j^{stor} \cdot S_{jpt}
 \end{aligned} \tag{8}$$

374 Here, R_{kp} are product- and consumer-specific revenues (€/t). The opportunity cost
 375 term penalises dereduced crop revenues in each year an AFS is established. Transport
 376 cost terms mirror the two-stage flow structure of the AFS supply chain with differenti-
 377 ated rates for raw and pre-treated biomass depending on the biomass type. Cost rates
 378 differ because as e.g. the density and water content of the shipped biomasses differ or
 379 because specialized equipment (e.g. for log transports) is used [REFERENCES]. More-
 380 over, site-specific establishment cost for planting the AFS trees is considered, next to
 381 yearly maintenance costs and harvesting cost. All these costs scale with the size of a
 382 site ($AREA_i$) measured in hectar. Maintenance cost comprise e.g. costs for trimming,
 383 replanting, and pruning unnecessary shoots. Storage costs are considered if biomasses are
 384 stored for more than a year to compensate direct costs (e.g. for storage site maintenance)
 385 but primarily opportunity cost due to potential material losses and capital commitment.
 386 The usual case is that AFS are harvested at the beginning of a year. Harvested biomass

dry during spring and summer such that the biomasses can be used in autumn or winter the same year. As drying is necessary for most industrial applications and beneficial for transport economics, drying storage in the year of harvest is not associated with additional cost.

3.4. Constraints

Constraints (9) limit each site to at most one establishment decision over the planning horizon. Constraints (10) enforce flow conservation in each period. If an activity in period $t \in \mathcal{T}$ takes place, there needs to be a preceeding activity (harvest or establishment) and a succeeding activity (harvest or termination). This way, (10) guarantee valid (possibly empty) activity paths over the complete time horizon for each site. Constraints (11) links the binary harvest-cycle variables to physical yield quantities via the age-dependent yield coefficients $\eta_{p(t-s)}$. These coefficients are pre-computed by evaluating the component-specific Gompertz functions (6) and (7) at stand age $a = t - s$ and scaled by site area AREA_i . Thus, implicitly we assume constant yields per hectare for each site i . Note that this simplification can be dropped and site-specific yields $\eta_{ip(t-s)}$ could be introduced without increasing numerical complexity of the AFS-SCD. Constraints (12) link harvest quantities to transport flows and ensure inventory balance at each facility, following the multi-period network-flow structure of OPTIMASS (De Meyer et al., 2014, 2016). Thereby, harvest quantities can be smaller than available AFS biomasses allowing partial harvesting. Constraints (13)–(14) enforce storage and processing capacity limits at each facility j . Constraints (15) implement product usage at multiple consumption sites k . Thereby, $(p', p) \in \mathcal{Q}$ defines a product mapping allowing high-quality product p' (e.g. stems) satisfying demand of a lower-quality products p up to maximum demand $D_{kpt}^{\max}$. This way, the cascading principle is operationalized as a central economic rationale.

$$\sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I} \quad (9)$$

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (10)$$

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq \sum_{(s,t) \in \mathcal{S}^{harv}} \eta_{p(t-s)} \cdot \text{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (11)$$

$$S_{jpt} = S_{jp,t-1} + \sum_{i \in \mathcal{I}} X_{ijpt} - \sum_{k \in \mathcal{K}, (p,p') \in \mathcal{Q}} X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (12)$$

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq \text{CAP}_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \quad (13)$$

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq \text{CAP}_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{harv} \quad (14)$$

$$\sum_{j \in \mathcal{J}, (p',p) \in \mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (15)$$

$$z_{ist} \in \{0, 1\} \quad \forall i \in \mathcal{I}, (s, t) \in \mathcal{S} \quad (16)$$

$$Y_{ipt}, X_{ijpt}, X_{jkp'pt}, S_{jpt} \geq 0 \quad \forall i, j, k, p, p', t \quad (17)$$

412 The yield coefficients η_{pa} in (11) are the critical bridge between the stand-level growth
 413 model of Chapter~2 and the supply chain optimisation. For stand age $a = t - s$ and
 414 product component p , they are computed as:

$$\eta_{pa} = f_p(a) \cdot M^{sl}(a) = \frac{h_p(a)}{\sum_{p'} h_{p'}(a)} \cdot a^\infty \cdot e^{-e^{-b'^1 \cdot (a-\tau)}}, \quad (18)$$

415 where $M^{sl}(a)$ follows from (5) and $f_p(a)$ follows from (7). For poplar clone Max~3 under
 416 SRAFS conditions, the component-specific Gompertz parameterisations from Section
 417 2.2 imply that $\eta_{stem,a}$ rises steeply between ages~4 and 10, while $\eta_{branch,a}$ and $\eta_{residue,a}$
 418 plateau earlier, as illustrated in Figure~??. This age-dependence of product fractions is
 419 the core mechanism through which rotation-length decisions — governed by $A^{\min}$ and
 420 $A^{\max}$ and the arc set $\mathcal{S}$ — determine achievable product portfolios and, hence, revenues
 421 in the objective function (8).

4. Case Study

The proposed MILP model is intended to support the design of agroforestry biomass supply chains for cleaner production in specific regional contexts. In a forthcoming case study, we focus on poplar-based alley-cropping systems in a temperate European setting, informed by empirical data from SRAFS trials in Southern Germany and SRC plantations in Central Europe (Huber et al., 2018; Trnka et al., 2008; Testa et al., 2014).

4.1. Study Region and Supply Chain Configuration

The proposed AFS-SCD model is instantiated on a real-world biomass supply chain located in the tri-state region of Saxony-Anhalt, northern Thuringia, and northern Saxony, Germany (Figure 4). This region combines the highest density of registered short-rotation coppice (SRC) Feldblöcke in eastern Germany with a cluster of major wood-processing industries that collectively represent one of the largest forest-based biomass demand centres in Central Europe. The spatial extent of the study region covers approximately 250×200 km, with road distances between potential KUP sites and consumer locations ranging from under 10 km to over 180 km.

The supply chain comprises three node types: (i) KUP production sites $\mathcal{I}$, (ii) pre-treatment and intermediate storage facilities $\mathcal{J}$, and (iii) consumer sites $\mathcal{K}$. Candidate KUP sites are drawn from the official Feldblock register of Saxony-Anhalt, filtered to parcels with a minimum area of 1 ha and classified as potentially suitable for SRC under the InVeKoS land-use scheme. Following the two-stage spatial clustering procedure described in Section-??, the $|\mathcal{I}| = 2239$ raw Feldblöcke are grouped into K super-sites using DBSCAN ($\varepsilon = 15$ km, $minPts = 3$) and Ward.D2 hierarchical clustering (maximum within-cluster radius 15 km), yielding a computationally tractable MILP instance while preserving the total eligible area. The HAC cluster assignments, visible as distinct colour zones in Figure 4, confirm a spatially coherent partitioning: dense KUP corridors along the Elbe and Saale floodplains form contiguous super-sites, while isolated parcels in the Harz foothills constitute smaller or singleton clusters.

Two types of intermediate infrastructure nodes are considered: conventional sawmills and logistics hubs. Sawmills perform primary mechanical processing (debarking, chipping) and can serve product streams P1 and P2; logistics hubs serve as consolidation

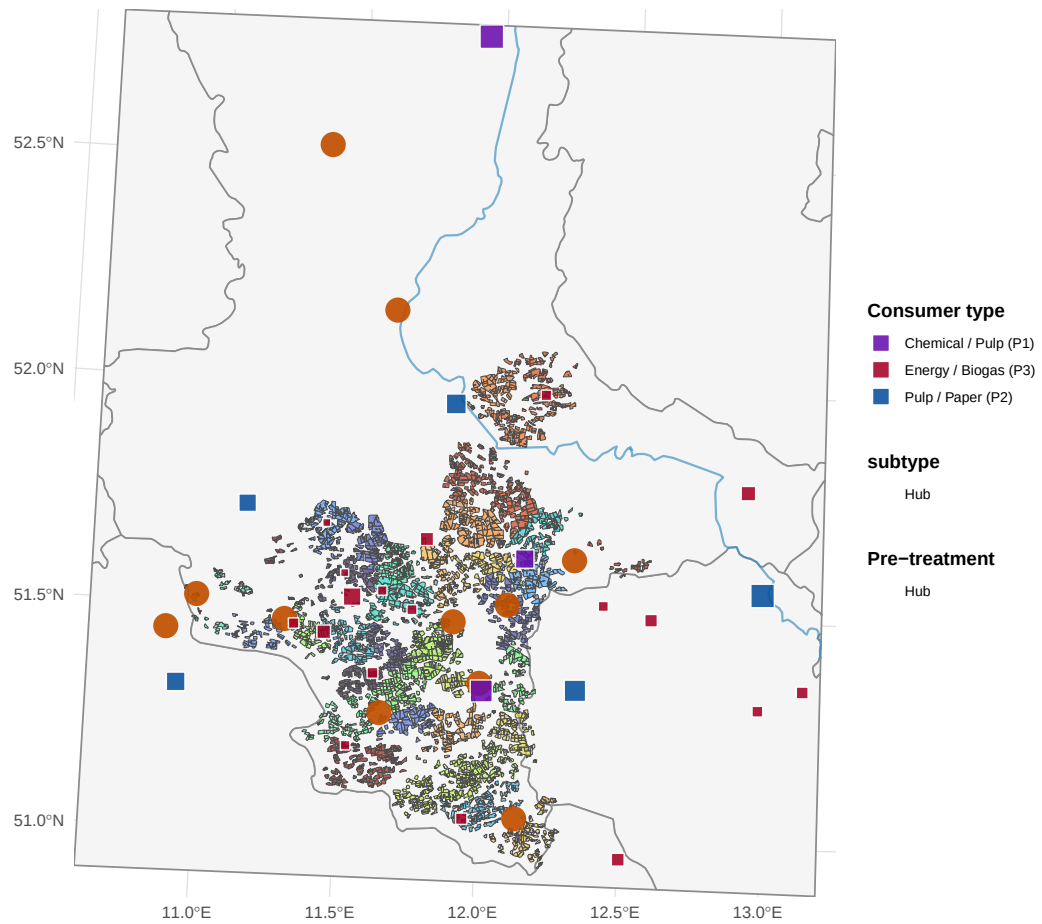


Figure 4: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential KUP/AFS sites grouped by HAC super-site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

and temporary storage points for all three product grades and are characterised by lower capital intensity but limited processing capacity. The $|\mathcal{J}| = 11$ candidate storage/pre-treatment locations are sourced from publicly registered forestry enterprises and existing logistic nodes in the study region. Road distances d_{ij} (site i to storage j) and d_{jk} (storage j to consumer k) are computed via the OSRM open-source routing engine using the German road network, providing realistic transport cost estimates that are embedded directly in the MILP objective.

4.2. Consumer Demand and Product Classification

Nine consumer sites are included in the base case ($|\mathcal{K}| = 9$), covering the full quality cascade from high-value chemical/pulp applications down to energy use (Table 4). Consumer demands $D_{kpt}^{\max}$ are derived from publicly reported annual throughput figures, production capacities, and regulatory filings, and are held constant across periods ($D_{kpt}^{\max} = D_{kp}^{\max} \forall t \in \mathcal{T}$) to reflect stable long-term offtake contracts.

Three biomass product grades are distinguished, reflecting the cascading-use principle embedded in the AFS-SCD objective:

- **P1 – Chemical/pulp grade** ($p = 1$): long-fibre hardwood chips suitable for chemical pulping and biorefinery conversion. Dominated by Mercer Stendal GmbH (Arneburg, 1{,}500 kt/yr) and Zellstoff Rosenthal/Mercer (Blankenstein, 1{,}000 kt/yr), which together account for 403% of total P1 demand. UPM Biochemicals (Leuna) contributes 500 kt/yr of high-purity beech-fraction demand for biochemical conversion. Gate prices for P1 range from 80 to 120 €/t wood input.
- **P2 – Sawlog/pulpwood grade** ($p = 2$): roundwood and chipped material suitable for mechanical processing in sawmills. Demand is distributed across six smaller facilities with individual annual demands of 5–300 kt/yr. Gate prices range from 55 to 75 €/t.
- **P3 – Energy/pellets grade** ($p = 3$): residual wood fractions (chips, bark, off-cuts) used for heat and electricity generation or pellet production. This grade absorbs material that cannot meet P1 or P2 quality thresholds, operationalising the cascade constraint in the MILP. Gate prices range from 30 to 40 €/t.

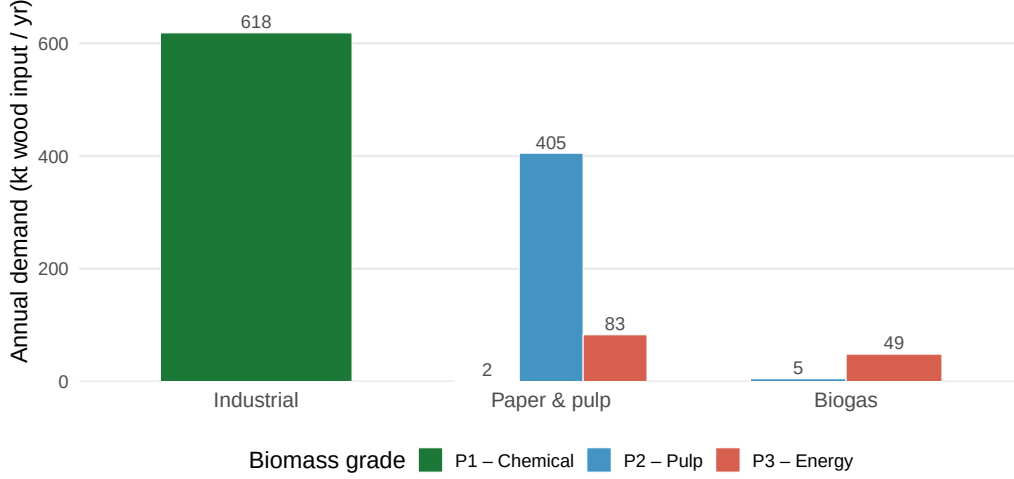


Figure 5: Total biomass demand by consumer type and grade (P1 chemical, P2 pulp & paper, P3 energy/biogas). Bars show aggregated annual demand in kt wood input per year across all individual plants within each consumer type.

481 The demand derivation follows a three-step procedure. First, publicly available ca-
 482 pacity data (kt wood input per year) are collected for each facility from official sources,
 483 industry reports, and corporate disclosures (Mercer Stendal GmbH, 2024; UPM Bio-
 484 chemicals, 2024). Second, the total annual capacity is assigned to the product grade that
 485 corresponds to the primary feedstock of each facility (P1 for chemical/biorefinery pro-
 486 cesses, P2 for mechanical sawmilling, P3 for thermal conversion and pellet lines). Third, a
 487 minor P3 component is added to sawmill facilities to capture by-product streams (bark,
 488 sawdust), consistent with observed industrial practice. The resulting demand matrix
 489 $D_{kp}^{\max}$ (kt/yr) is shown in Figure 5 and summarised in Table 4. Aggregate demand across
 490 the study region totals 620.5 kt/yr for P1, 410 kt/yr for P2, and 131.7 kt/yr for P3.

491 4.3. MILP Model Parameters

492 Table 5 summarises all scalar parameters of the AFS-SCD instance. The planning
 493 horizon covers $|\mathcal{T}| = 15$ periods, corresponding to one full poplar rotation (years 1–15),
 494 with a minimum harvest age $a^{\min} = 4$ and maximum productive stand age $a^{\max} =$
 495 15 (Lehmkuhl et al., 2023). Biomass yield functions η_{pa} (odt ha⁻¹ yr⁻¹) follow the
 496 age-structured Chapman-Richards profile calibrated in Section~??, where distinct yield

curves are assigned to each product grade to reflect the fraction of stem, branch, and foliage biomass that qualifies under each quality threshold.

Transport costs are disaggregated into two components reflecting the two-echelon logistics structure. Raw material transport from site i to pre-treatment facility j incurs a variable cost $c^{tr-raw} \cdot d_{ij}$ ($\text{€ t}^{-1} \text{ km}^{-1}$), while processed material transport from j to consumer k is costed at $c^{tr-pre} \cdot d_{jk}$. The pre-treatment rate c^{tr-pre} is set lower than c^{tr-raw} to reflect the higher load density and standardised packaging of chipped and pelletised material compared to loose roundwood. An opportunity cost parameter c^{opp} ($\text{€ ha}^{-1} \text{ yr}^{-1}$) accounts for foregone agricultural revenue on converted parcels, calibrated to the average arable land rental rate in Saxony-Anhalt (Statistisches Bundesamt (Destatis), 2023). Storage capacity at each facility j is bounded by $\bar{S}_j$, and processing capacity by $\bar{C}_j$ (kt yr^{-1}), both drawn from publicly available facility data.

4.4. Optimization results

5. Conclusion

6. Appendix

References

- Mohsen Arabi, Saeed Yaghoubi, and Hamed Tajik. Optimal design of a microalgae-based biofuel supply chain with co2 reduction and hydrogen production. *Computers & Chemical Engineering*, 130:106530, 2019. ISSN 0098-1354. doi: 10.1016/j.compchemeng.2019.106530.
- Marcos Barrio-Anta, Hortensia Sixto-Blanco, Isabel Cañellas-Rey de Viñas, and Fernando Castedo-Dorado. Dynamic growth model for short-rotation poplar plantations in northern and central-western Spain. *Forest Ecology and Management*, 255(3–4):1257–1268, 2008. ISSN 0378-1127. doi: 10.1016/j.foreco.2007.10.026.
- Claudia Cambero and Taraneh Sowlati. Assessment and optimization of forest biomass supply chains from economic, social and environmental perspectives—a review of literature. *Renewable and Sustainable Energy Reviews*, 36:62–73, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2014.04.041.
- Anouck De Meyer, Dirk Cattrysse, Jussi Rasinmäki, and Jos Van Orshoven. Methods to optimise the design and management of biomass-for-bioenergy supply chains: A review. *Renewable and Sustainable Energy Reviews*, 31:657–670, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2013.12.036.
- Anouck De Meyer, Dirk Cattrysse, and Jos Van Orshoven. A generic mathematical model to optimise strategic and tactical decisions in biomass-based supply chains (OPTIMASS). *European Journal of Operational Research*, 249(1):293–307, 2016. ISSN 0377-2217. doi: 10.1016/j.ejor.2015.08.003.

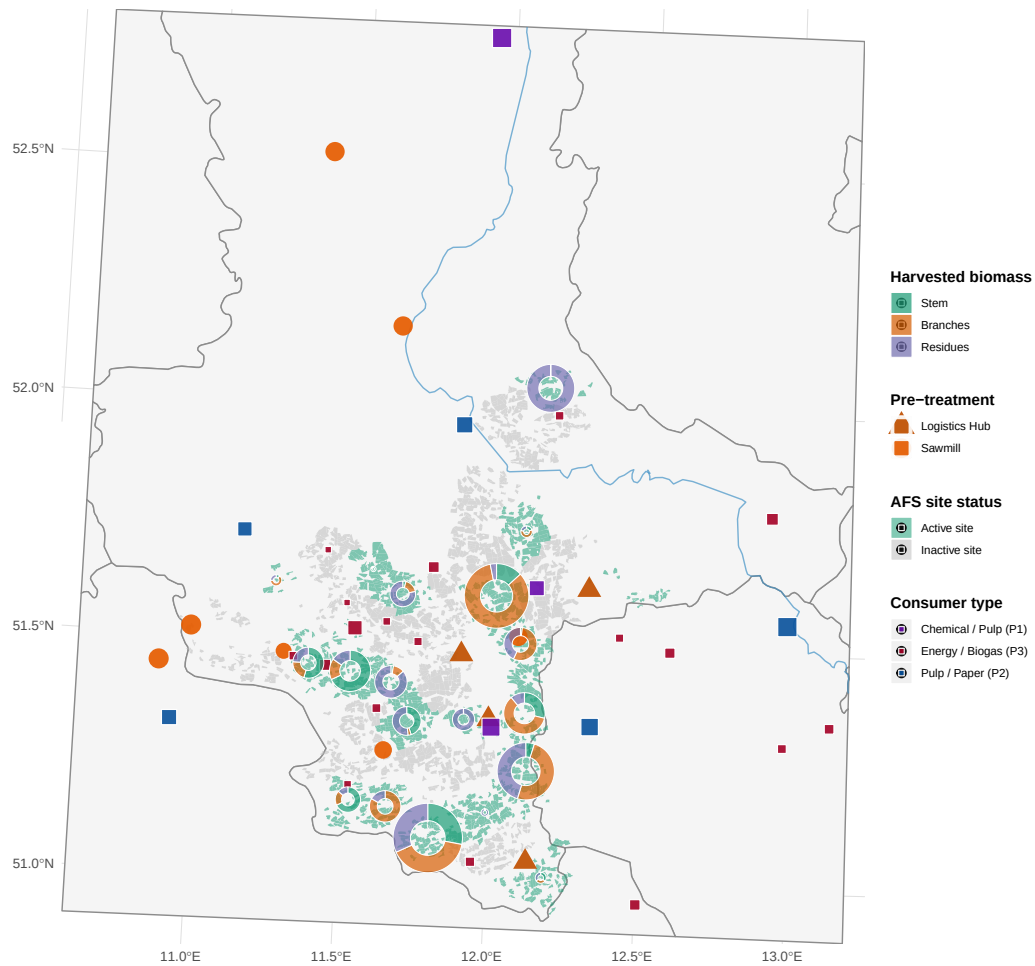


Figure 6: Optimal agroforestry supply chain network. Green polygons: active super-sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass (stem = P1, branches = P2, residues = P3), scaled by total harvested volume. Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub). Coloured squares: consumer sites by dominant demand category.

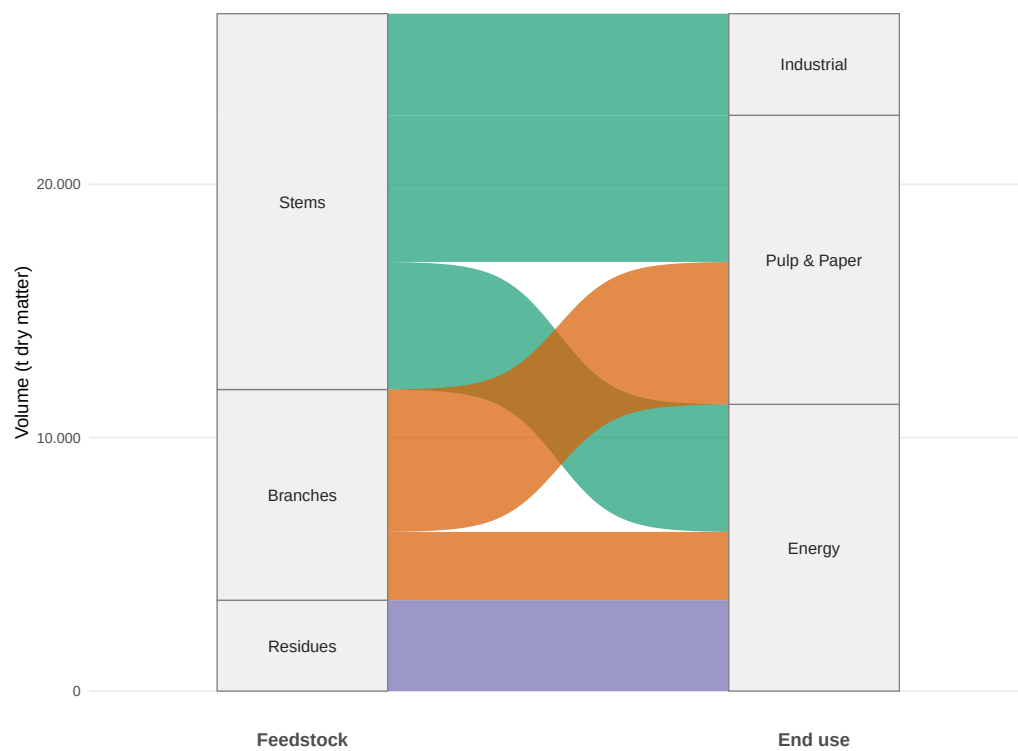


Figure 7: Total production quantity of biomass types and its industrial usage over planning horizon (in tons).

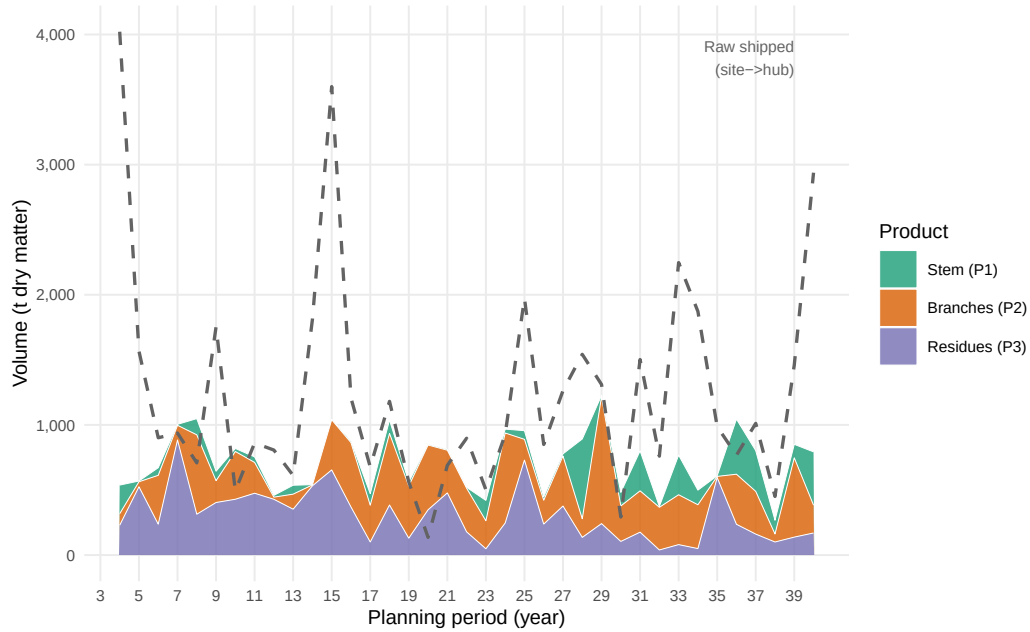


Figure 8: Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.



Figure 9: End-of-period inventory levels at storage/pre-treatment hubs, disaggregated by product type. Each panel corresponds to one active hub.

- Sandra Duni Ekşioğlu, Aditya Acharya, Lawrence E. Leightley, and Sumit Arora. Analyzing the design and management of biomass-to-biorefinery supply chain. *Computers & Industrial Engineering*, 57(4):1342–1352, 2009. ISSN 0360-8352. doi: 10.1016/j.cie.2009.07.003.
- Victor Espinoza, Selene Bautista, Patricia C. Narvaez, Mauricio Alfaro, and Mauricio Camargo. Sustainability assessment to support governmental biodiesel policy in Colombia: A system dynamics model. *Journal of Cleaner Production*, 141:1145–1163, 2017. ISSN 0959-6526. doi: 10.1016/j.jclepro.2016.09.168.
- Ralf J. Faasch and Gaia Patenaude. The economics of short rotation coppice in Germany. *Biomass and Bioenergy*, 45:27–40, 2012. ISSN 0961-9534. doi: 10.1016/j.biombioe.2012.04.012.
- Anil R. Graves, Paul J. Burgess, Fabrice Liagre, Andrea Pisanelli, Patrice Paris, Gerardo Moreno, Manuel Bellido, Manuela Mayus, Meindert Postma, Bernhard Schindler, Konstantinos Mantzanas, and Christian Dupraz. Development and application of bio-economic modelling to compare silvoarable, arable, and forestry systems in three European countries. *Ecological Engineering*, 29(4):434–449, 2007. ISSN 0925-8574. doi: 10.1016/j.ecoleng.2006.09.018.
- Henning Grunewald, Reinhold Böcker, B. Ulrich Schneider, Oliver Bens, and Reinhard F. Hüttl. Biomass production of black locust (*Robinia pseudoacacia* L.) short rotation plantations in Germany: growth and economics of different clones and management practices. *Bioresource Technology*, 98(2):281–286, 2007. ISSN 0960-8524. doi: 10.1016/j.biortech.2006.01.004.
- Georg Huber, Kurt-Jürgen Hülsbergen, and Harald Schmid. Allometric equations for above-ground tree-biomass estimation in Southern German short-rotation agroforestry systems. *Bioenergy Research*, 9(3):955–968, 2016. ISSN 1939-1234. doi: 10.1007/s12155-016-9744-z.
- Georg Huber, Christian Amann, Kurt-Jürgen Hülsbergen, and Harald Schmid. Yield potential of tree species in organic and conventional short-rotation agroforestry systems in Southern Germany. *Heliyon*, 4(1):e00543, 2018. ISSN 2405-8440. doi: 10.1016/j.heliyon.2018.e00543.
- Kaushalendra Kumar Jha. Biomass production and carbon balance in two hybrid poplar (*populus euramericana*) plantations raised with and without agriculture in southern France. *Journal of Forestry Research*, 29(6):1689–1701, 2018. ISSN 1007-662X. doi: 10.1007/s11676-017-0555-3.
- Patrick Lamers, Martin Junginger, Carlo Hamelinck, and André Faaij. Developments in liquid biofuels commodity trade. *Biofuels, Bioproducts and Biorefining*, 5(4):403–437, 2011. ISSN 1932-104X. doi: 10.1002/bbb.303.
- Frank Lehmkuhl, Alexander Stauch, Roman Schneider, Franziska Feil, and Henning Grunewald. Short-rotation coppice with poplar and willow in Germany: Biomass potentials, rotation lengths, and land-use interactions. *Biomass and Bioenergy*, 176:106908, 2023. ISSN 0961-9534. doi: 10.1016/j.biombioe.2023.106908.
- Krishna Teja Malladi and Taraneh Sowlati. Biomass logistics: A review of important features, optimization modelling and the new trends. *Renewable and Sustainable Energy Reviews*, 94:587–599, 2018. ISSN 1364-0321. doi: 10.1016/j.rser.2018.06.052.
- Babak Mansoornejad, Efstratios N. Pistikopoulos, and Paul R. Stuart. Integrating forest biorefinery metrics into supply chain design and analysis. *Forest Policy and Economics*, 26:43–54, 2013. ISSN

1389-9341. doi: 10.1016/j.forpol.2012.08.003.

Mercer Stendal GmbH. Mercer stendal: Zellstoffwerk Arneburg – kapazitäten und rohstoffversorgung. Unternehmenswebsite, 2024. URL <https://www.mercerint.com/de/stendal/>. Abgerufen: April 2026.

P. K. Ramachandran Nair. *An Introduction to Agroforestry*. Kluwer Academic Publishers, Dordrecht, 1993. ISBN 978-0-7923-2134-7.

Marcin Niemczyk and Brian R. Thomas. Does intensive management affect wood density and fiber properties of hybrid poplar grown in short-rotation plantation? *Forests*, 12(4):399, 2021. ISSN 1999-4907. doi: 10.3390/f12040399.

Bhavna Sharma, Robert G. Ingalls, Christopher L. Jones, and Ajay Khanchi. Biomass supply chain design and analysis: Basis, overview, modeling, challenges, and future. *Renewable and Sustainable Energy Reviews*, 24:608–627, 2013. ISSN 1364-0321. doi: 10.1016/j.rser.2013.03.049.

Raffaele Spinelli, Carla Nati, and Natascia Magagnotti. Using modified foragers to harvest short-rotation poplar plantations. *Biomass and Bioenergy*, 33(5):817–821, 2009. ISSN 0961-9534. doi: 10.1016/j.biombioe.2009.01.002.

Statistisches Bundesamt (Destatis). Kaufwerte und pachtpreise für landwirtschaftliche grundstücke 2022 – Fachserie 3 Reihe 2.4. Technical report, Statistisches Bundesamt, Wiesbaden, 2023. URL <https://www.destatis.de/DE/Themen/Branchen-Unternehmen/Landwirtschaft-Forstwirtschaft-Fischerei/Flaechennutzung/Publicationen/Downloads-Flaechennutzung/kaufwerte-landwirtschaft-2030240227004.pdf>. Pachtpreise Sachsen-Anhalt, Tabelle 3.1.

Riccardo Testa, Anna Maria Di Trapani, Mario Foderà, Filippo Sgroi, and Salvatore Tudisca. Economic evaluation of introduction of poplar as biomass crop in Italy. *Renewable and Sustainable Energy Reviews*, 38:498–506, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2014.07.007.

Eric Toensmeier. The carbon farming solution: A global toolkit of perennial crops and regenerative agriculture practices for climate change mitigation and food security. In *The Carbon Farming Solution*. Chelsea Green Publishing, White River Junction, VT, 2017. ISBN 978-1-60358-571-2.

Miroslav Trnka, Jana Fialová, Vladimír Koutecký, Milan Fajman, Zdeněk Žalud, and Milan Havlíček. Biomass production and survival rates of selected poplar clones grown under a short-rotation system on agricultural land. *Plant, Soil and Environment*, 54(2):78–88, 2008. ISSN 1214-1178. doi: 10.17221/2752-PSE.

UPM Biochemicals. UPM biochemicals leuna: Bioraffinerie für laubholz-biomassekonversion. Unternehmenswebsite, 2024. URL <https://www.upmbiochemicals.com/de/>. Abgerufen: April 2026.

Dajun Yue, Fengqi You, and Seth W. Snyder. Biomass-to-bioenergy and biofuel supply chain optimization: Overview, key issues and challenges. *Computers & Chemical Engineering*, 66:36–56, 2014. ISSN 0098-1354. doi: 10.1016/j.compchemeng.2013.11.016.

Sayed Mojib Zahraee, Nirajan Shiwakoti, and Peter Stasinopoulos. Biomass supply chain environmental, economic, and social sustainability: A systematic literature review. *Industrial Crops and Products*, 145:112144, 2020. ISSN 0926-6690. doi: 10.1016/j.indcrop.2020.112144.

Table 3: Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \dots, T_{\max}\}$	Planning periods (years)
$\mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$	Extended time set incl. dummy nodes
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \dots, T_{\max} - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T_{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}$	Termination arcs to dummy sink $T_{\max} + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment $\cup$ harvest $\cup$ termination)
Decision Variables	
$z_{ist} \in \{0, 1\}$	1 if s and t are consecutive harvest cycles at site i
$Y_{ipt} \geq 0$	Harvest quantity (t) of product p at site i in period t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
$T_{\max}$	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of product p at stand age a , derived from (5)–(7)
R_{kp}	Revenue (€/t) for product p at consumer k
C_i^{est}	Establishment cost (€/ha) at site i
C_i^{opp}	Annual opportunity cost (€/ha) at site i
C_i^{harv}	Harvest & refitting cost (€/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (€/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (€/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (€/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}}$	Storage capacity (t) at facility j
$\text{CAP}_j^{\text{proc}}$	Processing throughput capacity (t/period) at facility j
$D_{kpt}^{\max}$	Maximum demand (t) of product p at consumer k in period t

Table 4: Consumer demand quantities and gate prices by product grade. P1: chemical/pulp; P2: sawlog/pulpwood; P3: energy/pellets. Demands in kt wood input per year; prices in €/t wood input at the factory gate.

Consumer	Annual demand			Gate price		
	P1 (kt/yr)	P2 (kt/yr)	P3 (kt/yr)	P1 (€/t)	P2 (€/t)	P3 (€/t)
consumer 1	300.0	–	–	100	–	–
consumer 2	100.0	–	–	120	–	–
consumer 3	–	300	50.0	–	75	40
consumer 4	200.0	–	–	95	–	–
consumer 5	–	30	5.0	–	70	35
consumer 6	–	15	5.0	–	65	35
consumer 7	–	10	3.0	–	60	30
consumer 8	–	5	8.0	–	55	30
consumer 9	2.0	50	20.0	80	70	38
consumer 10	–	–	1.7	–	–	35
consumer 11	–	–	2.9	–	–	35
consumer 12	–	–	2.5	–	–	35
consumer 13	–	–	2.3	–	–	35
consumer 14	–	–	1.9	–	–	35
consumer 15	–	–	2.7	–	–	35
consumer 16	–	–	1.7	–	–	35
consumer 17	–	–	1.9	–	–	35
consumer 18	–	–	1.6	–	–	35
consumer 19	–	–	1.5	–	–	35
consumer 20	–	–	1.5	–	–	35
consumer 21	–	–	1.8	–	–	35
consumer 22	–	–	3.5	–	–	35
consumer 23	–	–	1.6	–	–	35
consumer 24	–	–	2.1	–	–	35
consumer 25	–	–	4.5	–	–	35
consumer 26	–	–	1.8	–	–	35
consumer 27	–	–	3.2	–	–	35
consumer 28	18.5	–	–	55	–	–

Table 5: AFS-SCD base-case parameter values for the Saxony-Anhalt case study. Transport costs refer to road distance (OSRM). Gate prices are weighted averages across consumer sites.

Parameter	Description	Base-case value	Unit
Index sets			
$ \mathcal{T} $	Planning periods	15	yr
$ \mathcal{I} $	KUP super-sites (post-clustering)	K (clustering-dependent)	–
$ \mathcal{J} $	Pre-treatment / storage nodes	11	–
$ \mathcal{K} $	Consumer sites	9	–
$ \mathcal{P} $	Product grades	3	–
Biomass dynamics			
$a^{\min}$	Minimum harvest age (years)	4	yr
$a^{\max}$	Maximum stand age (years)	15	yr
Cost parameters			
c^{tr-raw}	Raw transport cost	0.08	€ t ⁻¹ km ⁻¹
c^{tr-pre}	Pre-processed transport cost	0.05	€ t ⁻¹ km ⁻¹
c^{opp}	Opportunity cost (foregone arable rent)	280	€ ha ⁻¹ yr ⁻¹
Revenue parameters			
$\bar{R}_{P1}$	Mean gate price P1	99	€ t ⁻¹
$\bar{R}_{P2}$	Mean gate price P2	69	€ t ⁻¹
$\bar{R}_{P3}$	Mean gate price P3	36	€ t ⁻¹